HISSA

Tokenised co-lending settlement on NPCI DRUNIX

One shared ledger for banks and NBFCs that lend together. Repayments split by code the moment they land, one repayment status for both books, and a seat for the regulator.

hissa (Hindi) = share

One ₹10,000 loan, held as tokens



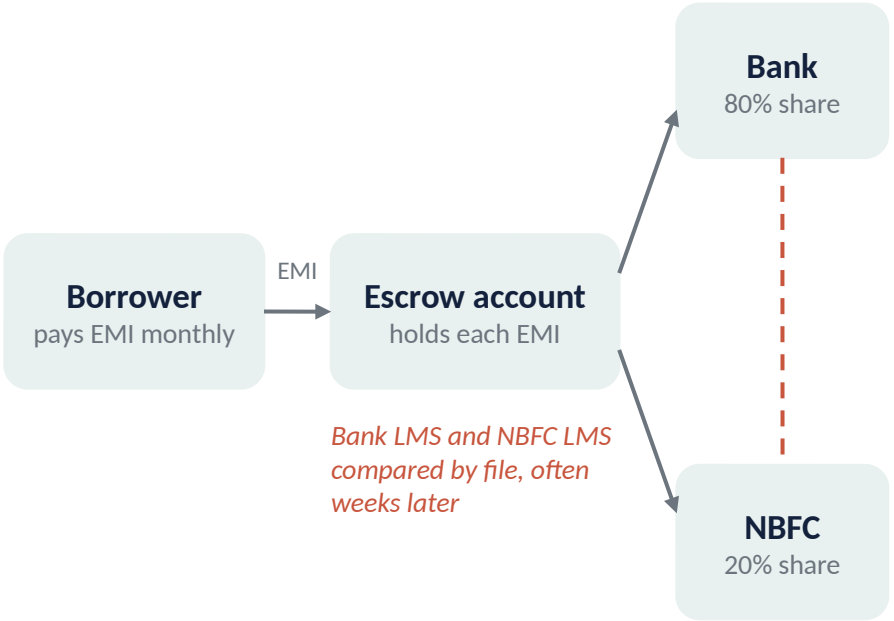
8,000 tokens: bank · 2,000 tokens: NBFC

Every EMI is split by these tokens, automatically, the second it arrives.

Team Prime Minds · Citi × NPCI DRUNIX Hackathon 2026 · Track: Financial Inclusion

Co-lending today: one loan, two notebooks

- A bank and an NBFC fund one loan together, usually 80:20. The NBFC finds and serves the borrower; the bank brings most of the money.
- Every EMI has to be split in that ratio and routed through an escrow account.
- Both lenders must record the same repayment status (on time, SMA, NPA) for the same borrower.
- Today each side keeps its own loan system. They match the two by exchanging files and spreadsheets after the fact.



Where it breaks: three pains, one root cause

1 Settlement

EMIs are split in daily or weekly batches. Files from one system are matched against the other and the misfits are fixed by hand. Money sits idle while this happens.

2 Classification

If the NBFC marks a loan late and the bank does not, the two report different risk for the same borrower. One failed file job and the books disagree.

3 Plumbing and oversight

Every bank and NBFC pair builds its own integration. Vendor tools serve one pair at a time. RBI sees nothing live and audits by sampling later.

Root cause: two regulated lenders keep separate books about one loan.

Why now: the rules changed on 1 January 2026

RBI Co-Lending Arrangements Directions, 2025

1. Each lender keeps at least 10% of every loan on its own books
2. All disbursements and repayments go through an escrow account
3. The partner's share must be taken within 15 days
4. A late or NPA mark must reach both books by the next working day
5. Co-lent loans come under audit

Every existing arrangement has to be re-platformed to meet these.

₹1 lakh cr

NBFC co-lending book, growing 35 to 40% a year

134%

growth in co-lending disbursements in FY24

1 in 3

NBFCs actually run co-lending; operations cost is the barrier

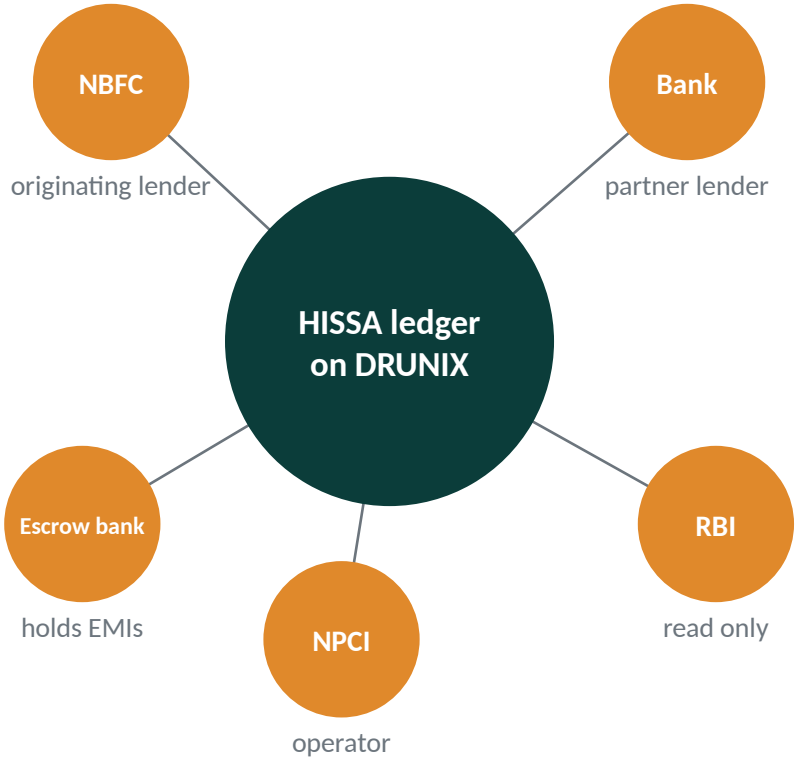
Jan 2026

the date every arrangement must comply by

Sources: Crisil Ratings (Apr 2024), Grant Thornton (co-lending in India), RBI Directions dated 6 Aug 2025

HISSA: NACH for co-lending

- A shared ledger that all five parties can read and none can quietly change.
- Each loan is cut into hissa (share) tokens, one token per rupee, held by each lender in its agreed ratio.
- When a repayment lands, code splits it the same second and both books update together.
- A late mark set once is seen by both lenders in minutes.
- Selling a share to another bank is a token transfer, not a re-papering exercise.
- RBI gets a read-only seat and can query the whole book in SQL.



How a loan lives on HISSA

1 Agreement

Ratio, rate, fee split, DLG cap and escrow details are registered once. Terms stay private to the two partners.

2 Tokenise

On disbursal, units are minted 80:20 to the two lenders. The 10% retention rule is enforced by code, not by audit.

3 Collect and split

UPI AutoPay or NACH brings the EMI in. The waterfall runs and both lenders are credited the same day, T+0.

4 Classify

DPD, SMA and NPA are one shared status with a worst-of rule. Set once, seen by both books in minutes.

5 Transfer

A lender sells part of its share to another bank by moving tokens, with the partner's consent. Securitisation later.

6 Supervise

RBI's read-only node runs SQL over the ledger. "All SMA-2 loans by NBFC" is one query, not a quarterly return.

A ₹5 lakh loan, start to finish

Day 0: Disbursal

₹5,00,000 goes out. Tokens minted: 4,00,000 to Bank A, 1,00,000 to the NBFC.

Month 1: EMI arrives

₹12,000 lands via UPI AutoPay. Waterfall runs: ₹9,600 to Bank A, ₹2,400 to the NBFC, the same second.

Month 4: EMI missed

DPD crosses 30. SMA-1 is set once on the ledger and both books show it in minutes.

Month 6: Share sold

Bank A sells 30% of its holding to Bank B. Tokens move; the next EMI splits three ways automatically.

Any day: RBI checks

RBI's node asks in SQL: every SMA-1 loan by NBFC, by vintage. Answer in seconds, no filing.

Why a shared ledger and not just a database

What each party needs	What a permissioned ledger gives
1 One record of the loan that no single party owns	A permissioned multi-organisation network run by a neutral operator, NPCI
2 No change unless both lenders agree	Endorsement policy: a transaction is valid only with both lenders' signatures
3 Proof for RBI that nothing was altered later	An append-only chain with a read-only regulator node
4 Fee deals and borrower details kept between partners	Private data collections shared only with the pair, hashes on the channel

A vendor database gives one or two of these. A multi-organisation ledger gives all four at once.

Built on DRUNIX, piece by piece

DRUNIX element	How HISSA uses it	In plain words
Lite Peer / Committing Peer split	Each org adds Lite Peers for endorsement on EMI peak days (1st to 10th); Committing Peers validate and commit	Extra cashiers on the busiest days, without copying the ledger
Stateless Validation Service	Scaled up for NACH presentation batches, down afterwards	A team of checkers that grows when the rush comes
YugabyteDB state database	RBI and escrow views run SQL straight on ledger state	A ledger you can search like a spreadsheet
KeyDB transient store, private data	Borrower identity and per-pair commercial terms in private collections; fewer network hops per write	A locked drawer only the right parties can open
Chaincode in Go	cla, participation-token, escrow, settlement-waterfall, classification, tle, dlg	The rulebook, written as code
Multi-organisation design	NBFC, bank, escrow bank, NPCI operator and orderer, RBI observer on one channel	Same shape as NACH: neutral operator, members keep their own books
Falcon	Helm-based deployment of the network on Kubernetes	The tool that sets up all the machines
NPCI APIs, Fabric Gateway SDK	UPI AutoPay and NACH events in; REST APIs and event listeners out	The pipes the money actually travels through

Architecture

Money rails: UPI AutoPay · NACH · RTGS (one netted instruction per bank per day)

Gateway: Node service on the Fabric Gateway SDK. REST APIs for lenders, collection webhooks in, chaincode event listeners out, four dashboards (NBFC, Bank, Escrow, RBI SQL).

DRUNIX network (NPCI's enhanced fork of Hyperledger Fabric)

NBFC	Bank	Escrow bank	NPCI	RBI
originating RE	partner RE	holds EMIs	operator, orderer	read-only observer

Chaincode (Go): cla · participation-token · escrow · settlement-waterfall · classification · tle · dlg

YugabyteDB state DB	KeyDB transient store + private data collections	Lite + Committing Peers, Validation Service	Falcon on Kubernetes

What changes

T+0

same-day settlement

from T+1 to T+3 batch cycles

Minutes

to sync a late or NPA mark

from next working day

0

reconciliation breaks

one state, one waterfall, by construction

₹200 cr+

idle money released

estimate: a 2-day lag on ₹3,500 to 4,000 crore of monthly collections on a ₹1 lakh crore book

1

onboarding per lender

instead of one per partner: N × M integrations become N

Compliance built in. The five RBI rules become properties of the ledger, not controls each lender has to build.

Inclusion. One onboarding opens bank balance sheets to the mid-size and small NBFCs that serve MSME, gold and affordable housing borrowers.

Figures marked estimate are ours. Market figures: Crisil Ratings, Grant Thornton.

Build plan: 10 to 22 November

Days 1 to 2	Days 3 to 6	Days 7 to 9	Days 10 to 11	Day 12
Four-org network	Chaincode + tests	Gateway + dashboards	Private data, load test, Falcon	Demo

Environment setup happens during the 26 Oct to 9 Nov evaluation window, so all twelve build days go to code.

What gets built

- Four-org DRUNIX test network
- Seven Go contracts with unit tests
- Node gateway; simulated AutoPay and NACH webhook
- Four dashboards: NBFC, bank, escrow, RBI SQL
- Private data collections per lender pair
- Load run on 2 Lite Peers and 2 Validation Service instances

Three-minute demo

1. Register a CLA: 80:20, blended rate, DLG cap
2. Disburse ₹5 lakh; tokens mint 80:20
3. EMI arrives by AutoPay; the split lands instantly
4. DPD slips; SMA-1 shows on both books
5. Bank A sells 30% to Bank B
6. RBI runs "all SMA-2 loans by NBFC" in SQL

Our moat, and what we need

Four reasons this is hard to copy once it is running

1. **Neutral by design** A utility that competitors and the regulator both trust cannot be copied by a pairwise vendor.
2. **Network effect** Every lender that joins makes the next integration unnecessary. N × M becomes N.
3. **Data gravity** The ledger becomes the canonical classification record for every co-lent loan.
4. **Timing** Built against a hard deadline, 1 January 2026, on rails NPCI already runs.

What we ask

- NPCI sandbox access for UPI AutoPay and NACH events
- A mentor who has run a co-lending or NBFC partnership desk at Citi or a partner bank
- Feedback on the token model, so it can be contributed back to DRUNIX

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